



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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CONVERSATION-STATE BASED APPROACH

Assignment on

*Assignment Report submitted in partial fulfilment of the requirements for the
Course of*

DSA0308 – Natural Language Processing

to the award of the degree of

BACHELOR OF Technology

IN

ARTIFICIAL INTELLIGENCE AND Machine Learning

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Hybrid NLP and Machine Learning Model

1. Problem Statement and Problem Formulation

Problem Statement

An e-commerce chatbot must understand customer messages even when they contain morphological variations, different word combinations, negation, ambiguity and references to previous conversation turns. A system based only on rules may become difficult to maintain, while a machine-learning model alone may incorrectly classify context-sensitive statements.

For example, in:

“I don't want to cancel it. I just want to know when it will arrive.”

the word *cancel* may strongly indicate Order Cancellation, but the complete sentence actually expresses a **rejection of cancellation** and a request for delivery information.

Therefore, a hybrid system is proposed that combines linguistic NLP processing, machine learning classification and rule-based contextual correction.

Problem Formulation

Input: Multi-turn customer conversation.

Output:

- Extracted entities
- Linguistic features
- Initial ML intent
- Contextual corrections
- Final intent
- Retrieved information
- Generated response
- Tamil translation

The target intents include Order Tracking, Order Cancellation, Product Information, Return Request, Refund Request, Payment Problem, Delivery Information, Account Problem, Complaint and General Inquiry.

2. Objective and Expected Outcomes

Objectives

The objectives are to:

1. Preprocess customer conversations.
2. Extract morphological and syntactic features.
3. Generate unigram, bigram and trigram features.
4. perform POS tagging and CFG analysis.
5. Detect negation.
6. Extract semantic features.
7. Resolve ambiguity and references.
8. Use an ML classifier for initial intent prediction.
9. Apply rule-based contextual correction.
10. Retrieve relevant information and generate a suitable response.
11. Translate the response into Tamil.

Expected Outcomes

The system should correctly identify the customer's requirement even when statistical and linguistic signals conflict.

For the given conversation, the expected final result is:

Final Intent: Order Tracking / Delivery Information

3. Requirements, Constraints and Assumptions

Requirements

The system should support:

- Text preprocessing
- Stemming
- N-gram generation
- POS tagging
- CFG parsing
- Negation detection
- Semantic feature extraction
- WSD
- Reference resolution

- ML-based classification
- Rule-based correction
- Information retrieval
- Response generation
- Tamil translation

These requirements correspond to the NLP pipeline and functional requirements defined in the assignment.

Constraints

- The solution must use multiple NLP concepts.
- Negated actions must not be treated as requested actions.
- Previous conversation context must be considered.
- Intermediate outputs should be available for validation.
- The system should not rely solely on keyword matching.

Assumptions

1. A labelled e-commerce intent dataset is available.
2. The chatbot processes conversations sequentially.
3. A predefined intent set is used.
4. A knowledge base is available for retrieval.
5. Translation is performed after response generation.

4. Application of Relevant Course Knowledge / Concepts

Morphology

Stemming handles variations such as:

arrive → arriv

arrived → arriv

tracking → track

cancelled → cancel

N-Grams and POS

The system uses unigrams, bigrams and trigrams to capture word relationships.

For example:

track order

when arrive

cancel my order

POS tags help distinguish actions, objects and other grammatical components.

CFG and Negation

The sentence structure is analyzed using CFG rules, while negation identifies statements such as:

don't want to cancel

as a rejected action.

Semantics and WSD

Semantic features help associate expressions such as *package*, *parcel* and *order* with the appropriate e-commerce concepts.

Reference Resolution

Expressions such as “it”, “this order” and “that item” are resolved using previous conversational information.

These concepts are part of the five-unit NLP coverage required by the assignment.

5. Design / Proposed Solution / Methodology

The proposed architecture is:

Customer Conversation



Regex



Morphology



N-grams + POS



CFG + Negation



Semantic Features



WSD + Reference Resolution



ML Classifier



Rule-based Context Correction



Final Intent



Information Retrieval



Response Generation



Tamil Translation

The **ML classifier** provides the initial prediction, while the rule-based layer checks whether that prediction agrees with linguistic and conversational evidence.

For example:

ML Prediction



Order Cancellation



Negation detected



Cancellation rejected



Delivery request detected



Final Intent = Order Tracking

6. Algorithm / Pseudocode / Flowchart

BEGIN

Load labelled dataset

Preprocess training data

Extract:

- morphological features

- n-grams

- POS features

- semantic features

Train ML intent classifier

FOR each conversation:

- preprocess message

- perform regex extraction

- perform stemming

- generate n-grams

- perform POS tagging and CFG parsing

- detect negation

- extract semantic information

- resolve ambiguous words

resolve references using conversation history

ML_intent ← classifier prediction

IF ML_intent is contradicted by negation:

 modify intent score

IF contextual evidence supports another intent:

 update intent score

final_intent ← highest corrected score

retrieve relevant information

generate response

translate response

END

The algorithm follows the assignment's requirement to combine linguistic processing, classification and contextual reasoning.

7. Implementation / Source Code and Environment / Tools Used

Component	Tool
Programming	Python
NLP	NLTK
Machine Learning	Scikit-learn

Component	Tool
Regex	Python re
Vectorization	TF-IDF
Classifier	Logistic Regression / Linear SVM
POS	NLTK
CFG	NLTK
State/Rules	Python
Translation	Translation library/API

Main Modules

preprocess()

extract_regex_features()

stem_words()

generate_ngrams()

pos_analysis()

cfg_parse()

detect_negation()

semantic_analysis()

resolve_reference()

predict_intent()

apply_context_rules()

generate_response()

translate_response()

The implementation uses a modular structure so that the ML component and linguistic correction component can be evaluated separately.

8. Test Cases and Expected / Actual Results

Test Case	Input	Expected Intent
TC1	“Where is my order?”	Order Tracking
TC2	“Cancel my order.”	Order Cancellation
TC3	“My package has not arrived.”	Order Tracking
TC4	“I want to return this product.”	Return Request
TC5	“My payment failed.”	Payment Problem
TC6	“I don't want to cancel it. When will it arrive?”	Order Tracking
TC7	“What features does this laptop have?”	Product Information

Critical Test

ML prediction:

Order Cancellation

Negation:

Detected

Context:

Customer rejects cancellation

Delivery request:

Detected

Final prediction:

Order Tracking

The actual confidence/probability values should be taken from the student's execution rather than inserted manually.

9. Execution Screenshots / Outputs

The implementation should capture screenshots showing:

Screenshot 1 — Preprocessing

```
1. PREPROCESSING / TOKENIZATION
-----
['i', 'was', 'expecting', 'the', 'phone', 'to', 'be', 'amazing', '.', 'the', 'display', 'is', 'bright', 'and', 'the', 'camera', 'takes', 'excellent', 'pictures', '.', 'but', 'the', 'battery', 'is', 'disappointing', '.', 'it', 'does', 'not', 'last', 'long', '.', 'and', 'i', 'am', 'unhappy', 'with', 'it', '.', 'however', ',', 'the', 'service', 'team', 'replaced', 'it', 'quickly', '.', 'which', 'was', 'impressive', '.', 'overall', ',', 'i', 'am', 'satisfied', 'with', 'the', 'phone', '.']
```

Screenshot 2 — Linguistic Features

```
2. FEATURES
-----
['i', 'was', 'expecting', 'the', 'phone', 'to', 'be', 'amazing', 'the', 'display', 'is', 'bright', 'and', 'the', 'camera', 'takes', 'excellent', 'pictures', 'but', 'the', 'battery', 'is', 'disappointing', 'it', 'does', 'not', 'last', 'long', 'and', 'i', 'am', 'unhappy', 'with', 'it', 'however', 'the', 'service', 'team', 'replaced', 'it', 'quickly', 'which', 'was', 'impressive', 'overall', 'i', 'am', 'satisfied', 'with', 'the', 'phone', '.']

4. BIGRAMS
-----
['i', 'was', 'expecting', 'the', 'phone', 'to', 'be', 'amazing', 'the', 'display', 'is', 'bright', 'and', 'the', 'camera', 'takes', 'excellent', 'pictures', 'but', 'the', 'battery', 'is', 'disappointing', 'it', 'does', 'not', 'last', 'long', 'and', 'i', 'am', 'unhappy', 'with', 'it', 'however', 'the', 'service', 'team', 'replaced', 'it', 'quickly', 'which', 'was', 'impressive', 'overall', 'i', 'am', 'satisfied', 'with', 'the', 'phone', '.']

5. TRIGRAMS
-----
['i', 'was', 'expecting', 'the', 'phone', 'to', 'be', 'amazing', 'the', 'display', 'is', 'bright', 'and', 'the', 'camera', 'takes', 'excellent', 'pictures', 'but', 'the', 'battery', 'is', 'disappointing', 'it', 'does', 'not', 'last', 'long', 'and', 'i', 'am', 'unhappy', 'with', 'it', 'however', 'the', 'service', 'team', 'replaced', 'it', 'quickly', 'which', 'was', 'impressive', 'overall', 'i', 'am', 'satisfied', 'with', 'the', 'phone', '.']
```

Screenshot 3 — ML Prediction

```
10. WORD SENSE DISAMBIGUATION
-----

Word: service
Possible senses: customer support, software service
Selected sense: customer support
Reason: The context mentions a service team replacing the phone.

Word: display
Possible senses: phone screen, public presentation
Selected sense: phone screen
Reason: The word occurs in the context of a phone.

Word: battery
Possible senses: phone battery, generic battery
Selected sense: phone battery
Reason: The review is explicitly about a phone.

11. REFERENCE RESOLUTION
-----

It in 'It does not last long' -> Battery
it in 'I am unhappy with it' -> Phone
it in 'replaced it quickly' -> Phone
which in 'which was impressive' -> Service/replacement action
```

Screenshot 4 — Rule Correction

```
3. NEGATION DETECTION
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Trigger: not
Scope: not last long ,
```

Screenshot 5 — Final Response

```
20. TAMIL TRANSLATION
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இந்த மதிப்பாய்வு ஒட்டுமொத்தமாக நேர்மறையானதாக வகைப்படுத்தப்படுகிறது. இரை மற்றும் கேமரா குறித்து நேர்மறையான கருத்துகள் உள்ளன. அதே நேரத்தில், பேட்டரி ஏமாற்றமளிப்பதாகவும் நீண்ட நேரம் நிற்கவில்லை என்றும் கூறப்படுவதால் அது எதிர்மறையான அம்சமாக உள்ளது. சேவை அணியினர் தொலைபேசியை விரைவாக மாற்றியதால் சேவை அனுபவம் நேர்மறையாக உள்ளது. இறுதியாக, பயனர் தொலைபேசியில் இருந்தியாக இருப்பதாகக் கூறுகிறார். எனவே, ஒட்டுமொத்த மதிப்பீடு நேர்மறையாகும்.
```

10. Results and Validation

Feature Validation

Component	Observation
Regex	Extracts structured information
Morphology	Reduces word-form variation
N-Grams	Captures local phrase patterns
POS	Adds grammatical information
CFG	Provides syntactic structure
Negation	Identifies rejected actions
WSD	Helps interpret ambiguous words
Reference Resolution	Connects expressions to previous entities
ML Classifier	Provides initial intent
Rule Correction	Resolves contextual conflicts

Main Conversation Result

Stage	Result
ML prediction	Cancellation-related
Negation analysis	Cancellation rejected
Semantic analysis	Delivery information requested
Reference resolution	“it” → order
Context correction	Changes prediction
Final intent	Order Tracking

For quantitative evaluation, the implementation should report actual **accuracy, precision, recall, F1-score and confusion matrix** obtained from the classifier. The assignment requires these forms of validation where applicable.

11. Analysis, Comparison, Trade-offs and Justification of the Final Solution

The major advantage of the hybrid approach is that it combines the strengths of two different methods.

Machine Learning

Advantages:

- Learns patterns from training data.
- Handles vocabulary variation.
- Reduces the need for manually writing every classification rule.

Disadvantages:

- May misinterpret negation.
- May depend strongly on training data.
- Can be difficult to explain for individual predictions.

Rule-Based Processing

Advantages:

- Easy to interpret.
- Useful for explicit linguistic conditions.
- Provides control over critical cases such as negation.

Disadvantages:

- Requires manual rule development.
- Large rule sets can become difficult to maintain.
- May not generalize well to unseen expressions.

Proposed Hybrid Model

Approach	Flexibility	Interpretability	Context Handling
Rule-based	Low	High	Moderate
ML	High	Moderate/Low	Limited

Approach	Flexibility	Interpretability	Context Handling
Hybrid	High	High	High

The hybrid solution is therefore justified because the ML model handles general language patterns while rules correct predictable contextual errors.

12. Broader Considerations / SDG Relevance

The chatbot relates to **SDG 9 – Industry, Innovation and Infrastructure**, as it applies NLP and machine learning to automated digital customer-service infrastructure.

Important considerations include:

- **Privacy:** Customer and order information should be protected.
- **Bias:** The classifier should be tested with varied language.
- **Transparency:** Rule-based corrections make important decisions easier to explain.
- **Safety:** Actions such as cancellation should require explicit confirmation.
- **Accessibility:** Tamil translation can improve regional-language accessibility.

13. Conclusion, Limitations and Possible Improvements

Conclusion

A hybrid NLP chatbot was proposed by combining linguistic processing, machine-learning classification and rule-based contextual correction.

The ML model provides a flexible initial intent prediction, while linguistic rules check negation, semantic evidence and conversational context. In the example conversation, the model may initially identify cancellation-related vocabulary, but the correction layer recognizes that cancellation is rejected and that the customer actually wants delivery information.

Therefore, the final intent is **Order Tracking / Delivery Information**.

Limitations

- Requires both training data and manually designed rules.
- Rules may need updating for new conversational patterns.
- ML accuracy depends on dataset quality.
- Complex references may still be difficult to resolve.

- Maintaining both components increases system complexity.

Possible Improvements

Future versions could use transformer-based classifiers, advanced coreference resolution, larger multilingual datasets and confidence-based human-agent escalation.

14. Individual Contribution of Group Members

The following should be modified according to the group's **actual contributions**.

Member	Contribution
R.Kalyan Kumar Reddy	Dataset preparation and preprocessing
M.Jagadeesh	Feature extraction and ML classifier
A Syed Thoufiq	CFG, negation and semantic processing
S Mohammed Abrar	WSD, reference resolution and contextual rules
V. Nagendra	Response generation, translation, testing and documentation
M. Sai Teja	Testing and validation

15. References

References should include only the resources actually used, such as:

1. NLP course textbook.
2. NLTK documentation.
3. Scikit-learn documentation.
4. Python documentation.
5. Research papers related to intent classification.
6. Dataset source and documentation.
7. Translation library/API documentation.
8. Software and development tools used.
9. AI-assisted tools, if used.

The assignment explicitly requires appropriate references for datasets, software/tools, research sources and AI-assisted tools where applicable.

16. One-Page Individual Reflection

Individual Reflection

Working on the Conversation-State Based Approach helped me understand how different Natural Language Processing techniques can work together to build an accurate and context-aware conversational system. My main contribution to the project was Word Sense Disambiguation (WSD), Reference Resolution, and Contextual Rules. Through Word Sense Disambiguation, I learned how the meaning of an ambiguous word can be identified based on its surrounding context. This is useful because the same word can have different meanings in different situations. I also worked on Reference Resolution, which helps the system understand expressions such as “it,” “this order,” and “that item” by connecting them with entities mentioned earlier in the conversation.

Another major part of my contribution was implementing contextual rules to improve the initial prediction made by the machine-learning classifier. These rules check the prediction using linguistic and conversational information. For example, in the sentence “I don't want to cancel it. When will it arrive?” the word *cancel* may cause the ML model to predict Order Cancellation. However, the context indicates that the customer is rejecting cancellation and asking about delivery. The contextual rules help correct this prediction to Order Tracking / Delivery Information. One of the main challenges I faced was combining WSD, reference resolution, and contextual information with the machine-learning component. This helped me understand that a chatbot cannot always depend only on keywords or ML predictions. Contextual information from previous conversation turns is important for correctly understanding the customer's actual intention.

I also learned that a hybrid approach provides better flexibility and control. Machine learning can identify general language patterns, while rule-based processing can handle specific conditions and contextual conflicts.

Overall, this assignment improved my practical understanding of semantic analysis, WSD, reference resolution, conversation context, and rule-based reasoning. It showed me how different NLP techniques contribute to a reliable chatbot. In the future, I would improve the system using stronger reference-resolution and semantic models and test it with larger and more diverse conversational datasets.